Finite-Lifetime Single-Port Product Capture and Context-Safe Weight Broadcast for Event-Driven Spiking Optical Flow

Authors to be inserted

***Abstract*—Low firing activity does not automatically accelerate event-driven optical flow: repeated arithmetic and weight delivery remain constrained by finite storage, ports, and accumulator ownership. This brief presents two exact reuse islands for a frozen spiking optical-flow workload. Both decide reuse before the resource being suppressed and preserve private completion state. C1 captures a product only while an exact-subset parent remains resident in a single-1RW store, issues the residual, suppresses dead writes, and retires completion atomically. On 51.84 million source rows, its same-ledger cycle model reduces component time by 40.99% (1.6945×) versus strongest-zero skipping; a mapped 28-nm nine-SRAM island occupies 166,514 μm^2 and meets 3-ns setup/hold. C2 shares one Acc24 fabric across eight typed signed sources. Token-set broadcast (TSBG) reuses only weight delivery while keeping products, destinations, and accumulators private. Across 2,880 fixed real-activity workloads spanning all 12 FC1 and 12 FC2 layer identities, same-port/cache VCS reduces post-load cycles by 45.49% (1.8345×) and scalar weight reads by 58.13%, with 0.0118% matched logic-area overhead. Against equal-bandwidth K1×8, K8 uses 77.61% less logic and provides 4.541× directed-throughput/logic-area. Results remain component-level and prelayout; no whole-network speedup is inferred.**

***Index Terms*—Digital accelerator, event-based vision, optical flow, product sparsity, single-port SRAM, spiking neural networks.**

EVENT cameras make inactivity visible, but sparse arithmetic is only one term in execution time. In a binary-event optical-flow deployment, nonzero sources still trigger parent-state reads, residual issues, and psum writes. A design that reports only operation sparsity can therefore be slower than an equal-resource zero-skipping baseline once the parent working set is finite and the store is single-ported.

Prosperity [1] and Phi [2] show that useful SNN speedups come from executable reuse or patterns, not firing rate alone. Digital neuromorphic processors such as ODIN [3] and event-stream accelerators such as SpinalFlow and SNE [4], [5] establish spike-driven datapaths. TCAS-II SRAM compute-in-memory macros exploit sparsity inside the array [6], [7]. Our object is different: repeated *products* of a convolution-like layer, subject to a 240-KiB-class parent lifetime and a physical 1RW contract.

This brief addresses two costs that remain after zero-source skipping: retaining reusable products and repeatedly delivering the same weights. C1 captures products only while

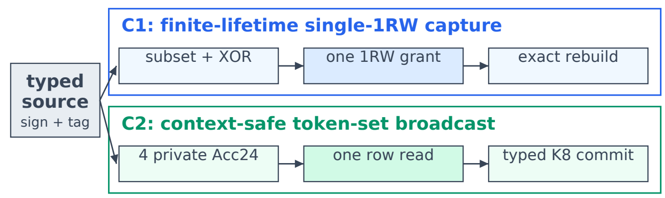

Fig. 1. Two exact reuse objects under one typed-source protocol. C1 retains only legal live parents in a finite 1RW store; C2/TSBG broadcasts one delivered weight row while preserving private signed products and Acc24 contexts.

their parents are live, schedules reads and writes through one physical port, and suppresses writes without a future consumer. C2 shares an Acc24 fabric across eight typed sources; its TSBG frontend groups consumers before weight reads so that one delivered row serves several contexts. Signs, destinations, products, and accumulators remain private. The common design principle is to decide reuse before incurring the access or computation it eliminates. The frozen Motion checkpoint supplies the workload; neither circuit requires retraining or changes to its source descriptors.

I. Workload and Arithmetic Model

The frozen Motion C12 ep34 checkpoint obtains 1.199514 average endpoint error (AEE) on an 825-frame local DSEC validation set and has a 5.6709% global firing rate. Hardware evaluation consumes the sealed ep34 source descriptors generated by fixed normal inference. Captured ATLIF sources entering the evaluated operators are binary; signed source codes represent downstream polarity and correction traffic rather than analog tensor values. The frozen software path uses Motion-XOR scoring with α=0.125 and K as its value carrier, while hardware quantization is disabled on that path. Model, VCS, and physical evidence remain explicitly tagged and are never multiplied.

Bottleneck convolution exposes repeated products only while parent and destination state remain resident. That invariant defines C1. Attention and decoder traces remain part of the frozen identity but are not promoted to performance contributions.

II. C1 Microarchitecture

A. Directory and Legal Parents

C1 first builds a ping-pong directory for each 64-row, 16-source mask task. A row may select only an earlier exact-

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The authors are with Huazhong University of Science and Technology, Wuhan 430074, China. TCAS-II is single-blind: replace this block with legal names, IEEE membership, e-mail, ORCID, and funding before submission.

subset row as its parent. Stable population-count ordering makes that choice unique: among legal parents the engine prefers the earliest still-live candidate so that later rows keep a deterministic residual. A live-parent bitmap records which completed rows remain observable. When the bitmap shows that no subsequent row can read a stored psum, that location is a dead write and is elided exactly rather than being scheduled as a drain.

The 240-KiB-class working set is the capacity that must be respected, not a DRAM-backed scratchpad. Nine 128×128 1RW SRAM leaves operate in parallel as one 128×1152 parent-product scratch and retain 96 signed 12-bit product lanes. Masks and the live-parent directory are control state; the downstream psum store is a separate boundary. The engine never assumes the ideal simultaneous 1R1W access that produced a higher but unphysicalized reuse ceiling. This is the object difference from an unbounded product-sparsity opportunity: reuse must survive capacity, port, parent lifetime, and completion constraints.

B. Residual Issue and Exact Reconstruct

A selected parent mask $\mathcal{M}_p \subset \mathcal{M}_r$ yields a disjoint residual. C1 issues only that residual and reconstructs the 96-lane signed product row by adding the stored parent:

$$\mathbf{P}_r = \mathbf{P}_p + \sum_{s \in \mathcal{M}_r \setminus \mathcal{M}_p} \mathbf{p}_s. \tag{1}$$

Exactness follows from the disjoint union, not from value prediction. If no resident legal parent exists, C1 selects the empty parent and issues the original source row. Sign, destination, and accumulator identity are never coalesced across rows. Thus sparsity changes delivered work without changing the operator result.

C. Single-Port Arbiter and Atomic Commit

The execution side combines one earliest-parent lookahead, a two-entry reserved response queue, and forwarding under a deadline-aware 1RW arbiter. A parent read, a residual product, and a psum write compete for the same macro port. The arbiter grants the access that would otherwise miss its row-completion deadline; the reserved queue absorbs one already-accepted response so that a later grant cannot orphan it. Forwarding supplies a just-written parent to the next residual without a second macro read when the bitmap still marks that parent live.

Final psum write and row completion are a single atomic event. Backpressure may change service order, but it cannot expose a partially reconstructed row: the typed-source terminal bit and the completion pulse retire together. A directed VCS tile on the first 64 real support masks of the cycle ledger is used later only to calibrate these event counts, not to replace the same-ledger cycle model.

Speculative parent hits are disallowed. A candidate is legal only if its mask is an exact subset and the live-parent bitmap is still set at grant time, not at directory build time. That second check is what makes (1) exact under eviction: a parent that died between directory construction and issue is treated as missing, and C1 falls back to the empty parent. The fallback is slower than a stale hit and is the price of never reconstructing from a psum that the store no longer owns.

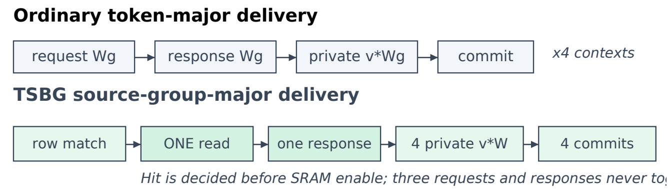

Fig. 2. TSBG suppresses redundant SRAM transactions before request. The returned row fans out to private signed products and Acc24 contexts; component ratios from different evidence classes are not multiplied.

III. C2 Typed-K8 and Context-Safe Broadcast

The C2 fabric accepts up to eight nonzero typed sources and shares the Acc24 array, endpoint, and control rather than replicating eight K1 engines. Its fair equal-service baseline is therefore K1×8, not one K1 lane. Eight independent memory-bank channels may return out of order; epoch, slot, generation, tag, slice, and source-channel fields bind every response before it can enter the signed bridge. A stale or misrouted response is rejected rather than accumulated.

TSBG changes only weight delivery. Four token contexts expose a common source-group row, so one fetched row can serve four independent signed products and four Acc24 contexts. For shared row W_g,

$$\text{fetch}(W_g) = 1, \qquad A_c \leftarrow A_c + v_c W_g, \quad (v_c, d_c, A_c) \text{ private.} \tag{2}$$

It is incorrect to share the product because equal source indices can carry different signs or destinations. Both ordinary and broadcast modes elaborate the same 192-bit row-live bitmap, four-row cache, public memory ports, and hierarchical priority encoder; only the static scheduling order changes from token-major to source-group-major. A cache hit is resolved before the bank request state, so a reused row suppresses the memory request and all downstream response traffic instead of merely gating the multiplier after a read.

Exact fallbacks are explicit. If token contexts disagree on a weight-row identity, C2 emits separate fetches. If C1 has no resident legal parent, it selects the empty parent and issues the original source row. Backpressure may alter service order but not ownership: terminal and atomic commit retire only after the corresponding private accumulator is complete.

Figure 2 makes the saved circuit event explicit. Ordinary token-major execution resolves the same row separately for each context. TSBG first proves a common row identity, then asserts one SRAM request and fans out the returned row. The other three read-enables, addresses, and responses are never generated; this is memory-traffic suppression, not a multiplier gate applied after the access. Product formation and retirement remain four private operations.

IV. Evaluation

All physical results use a commercial 28-nm flow at 3.0 ns. Reported points are prelayout, use ideal clocks and ZeroWireload, and have no extracted parasitics. C1 area includes

TABLE I
C1 SAME-LEDGER CAPTURE LADDER (51.84 M SOURCE ROWS).

Execution point	Cycles (M)	vs. zero	Boundary
Strong zero	648.741	1.000×	baseline
Same-coordinate bit	646.619	1.003×	baseline
C1 finite 1RW	382.849	1.6945×	cycle model
Concurrent-access	341.058	1.902×	ceiling only

TABLE II
TSBG-B4 FIXED 40-SAMPLE VCS SCOPE LADDER.

Scope	Loads	Ordinary	TSBG	Speedup
G48: FC1+4 FC2	1,920	12.523 M	5.124 M	2.4438×
G96/192: 8 FC2	960	80.129 M	45.381 M	1.7657×
all identities	**2,880**	**92.652 M**	**50.505 M**	**1.8345×**

nine generated SRAM leaves; C2 comparisons are logic-only and therefore remain separately labeled in Table IV.

Table I exposes the baseline ladder instead of attributing all inactivity to product capture. The ledger is 51.84 million source rows from ten `zurich_city_09_a` ep34 samples. Per-sample ratios span 1.661–1.723×, with a 1.6946× geometric mean. Strongest-zero skipping still walks parent-state traffic for every nonzero source; same-coordinate bit skipping on this ledger removes almost none of that traffic (1.003×). Exact parent reuse is what creates the material reduction. The remaining 12.25% gap from C1 to the concurrent-access ceiling is the tax of the physical single-port contract. We do not promote the ceiling as an attainable RTL result.

The cycle model charges the same resources the RTL would charge: one 1RW grant per accepted macro access, residual issue, forward, deadline hold, and atomic commit. It is not a CPU-time measurement and is not an RTL speedup. Only a future campaign that closes model, VCS, and DC/PT on one full ledger may use that stronger label.

The mapped nine-SRAM island occupies 166,514 μm^2. Mapped synthesis and prelayout PrimeTime use the foundry 28-nm standard-cell library and the nine compiler SRAM abstracts; no Yosys or educational library is used as an ASIC PPA proxy. PrimeTime reports setup/hold slack of +27.9/+1.8 ps at 3 ns under ideal clocks. Formality passes 16,549 mapped-to-mapped compare points on that netlist pair. In a separately bounded 64-row, 253-cycle directed window the island dissipates 29.08 mW and 22.07 nJ. That energy is a mixed-corner, prelayout PrimeTime PX estimate without SPEF and without a frame-level activity factor; it is not energy per DSEC frame and is not a chip measurement.

A separately sealed ep34 tile uses the first 64 real support masks of the cycle ledger. VCS matches 196 accepted issues, 58 parent edges, 31 dead-write elisions, 54/33 macro reads/writes, four forwards, six deadline holds, 14 stalls, and 64 commits/completions. Lane values are deterministic synthetic signed-12-bit data with zero prior psum. This calibrates modeled event counts and directed arithmetic; it does not itself produce the 1.6945× cycle ratio. Protocol errors, X-pessimism on energy, and failed sibling attempts in the same repository are not cited.

The 64-row task is the scheduling quantum that matches the ping-pong directory, not a claim that the deployed convolution is only 64 rows deep. Each task carries up to 16 typed sources and writes a 96-lane signed product. The nine macros are addressed as a single 1RW parent store: one grant moves either a mask, a psum, or a residual payload. Capacity eviction follows the live-parent bitmap rather than LRU of raw addresses, because a recently written parent may already be dead while an older parent remains legally observable. That replacement rule is why the concurrent-access ceiling in Table I is not a drop-in RTL mode: it would require a second port that the mapped island does not have.

The same-ledger model therefore reports two numbers that must not be conflated. 1.6945× is C1 versus strongest-zero on the physical 1RW contract. 1.902× is the unphysicalized ceiling. The 12.25% gap is disclosed so that a reader cannot mistake the ceiling for a second reported architecture.

For numerical binding, eight retained Conv3x3/ConvTranspose weights use per-output-channel narrow-range INT8 with a power-of-two scale. The largest absolute-code bound is 200,219, below Acc24; 1,280 Python-integer probes match and the observed final range is $[-29{,}680, 27{,}619]$. On 825 local DSEC frames the hardware-order mixed-precision candidate obtains 1.197367 AEE versus a 1.199514 historical baseline, passing the predefined +0.02 gate. Different GPU backend flags make this a compatibility check, not a causal accuracy improvement.

TSBG is evaluated on 2,880 fixed ep34 workloads selected independently of performance: 40 samples from four DSEC sequences, all 12 FC1 and 12 FC2 layer identities, and first/middle/last aligned B4 token quartets. The G48 engine handles all FC1 and four FC2 identities directly; the remaining six G96 and two G192 FC2 identities execute as functionally exact 48-source continuations. Activity comes from the frozen checkpoint, whereas arithmetic uses deterministic directed INT8 verification weights rather than captured model weights. Naturally nonzero descriptors in this population carry +1; directed negative-source tests separately cover the signed protocol. The G48 population is one continuous simulator-image lineage: 1,917 rows precede a host-orchestration successor that emits the final three, after which every row is parsed and independently sealed. A separate one-compile, one-simv continuation batch covers 960 workloads. Both use identical public ports and cache capacity; Table IV reports post-load execution. The G48 subset spans 2.381–2.546× by sequence. The continuation population contains 554 improvements, 404 ties, and two regressions of one and seven cycles (worst 0.99987×); these tails remain in the ratio of sums.

To test whether the three fixed B4 regions hide a position effect, we calibrate an independent cycle recurrence against all 2,880 VCS workloads and then replay every aligned B4 quartet in the same frozen 40-sample capture. Table III reports this model separately from RTL execution. Across 11.16 million quartets, TSBG reduces modeled cycles from 313.604 to 150.234 billion (2.0874× ratio of sums, or 52.09%) and scheduled scalar weight-read requests from 192.483 to 67.992 billion (64.68%). The four sequence ratios span only 2.0807–2.0968×; FC1 and FC2 obtain 2.1766× and 1.9005×, respec-

TABLE III
ALL-ALIGNED-B4 ROBUSTNESS [VCS-CALIBRATED CPU MODEL].

Scope	Ordinary cycles	TSBG cycles	Ratio
All 24 FC layers	313.604 G	150.234 G	2.0874×
FC1 / FC2	–	–	2.1766 / 1.9005×
Four sequences	–	–	2.0807–2.0968×
Observed-residual envelope	–	–	2.0870×

TABLE IV
ADMITTED COMPONENT EVIDENCE; OPEN CLAIMS ARE NOT INFERRED.

Class	Result	Boundary
C1	1.6945×; −40.99% time vs. strong zero[model]	Ten-sample same-ledger model; not RTL/system speedup
K8	1,913/1,945 cycles; 4.541× throughput/logic-area; logic −77.61%[VCS][DC]	vs. equal-bandwidth K1×8; five directed loads; logic-only
TSBG	92.652/50.505 M cycles (1.8345×); reads −58.13%[VCS]	2,880 real-activity loads; all 12 FC1/FC2 identities; fixed three regions, not full FC
C1 phys.	166,514 μm^2; +27.9/+1.8 ps; 29.08 mW / 22.07 nJ[DC/PT]	Nine-SRAM prelayout island; mixed-corner directed window; no SPEF
C1 store	18,432 B / 9 macros; 214,912 B / 105 macros / 0.988 mm^2[model]	Parent scratch integrated above; complete ledger is an area model, not integrated PPA

tively. Per-quartet p10/p50/p90 are 1.0000/1.5414/2.4388×. Because the mode is elaboration-time, there is no dynamic fallback: 3.10% of quartets are marginally slower and the worst case is 0.99755×.

For G≤48, the recurrence matches 3,840 ordinary/TSBG VCS cycle fields with zero fitted residual. G96/G192 comprises 780,000 quartets, or 6.981% of the full population; 779,040 (99.877% within this stratum) use the median wrapper residual from 960 keyed VCS anchors. Replacing all such ordinary/TSBG residuals by their observed minimum/maximum changes the aggregate only from 2.087430× to 2.086979×. This is an observed-envelope sensitivity, not a formal bound. Inputs retain real ep34 activity/sign descriptors, while arithmetic weights remain directed timing values; the result is a VCS-calibrated FC component model, not RTL, same-area, or whole-network execution.

Pre-read causality. A separate directed VCS ablation compares ordinary token-major execution, group-major post-read gating, and group-major pre-read TSBG. The post-read control performs the same scheduling but still requests, receives, and identity-checks the redundant weights before discarding them. The three axes issue 2,304/2,304/576 scalar reads, respectively, while each completes 4,608 signed products and 24 commits with zero mismatch. Thus the 75% request reduction in this directed case occurs before SRAM access; reordering followed by post-read gating does not produce it. This isolates request causality, not matched area or energy, and is separate from the 2,880-workload result.

The matched ordinary/TSBG logic ablation occupies 249,710/249,740 μm^2 (+0.0118%) and both axes meet 3-ns setup. Both retain a −16.4-ps prelayout hold diagnostic; consequently this brief does not claim a hold-closed C2 frequency. The common 288-KiB, 16-macro weight capacity is a 558,507-μm^2 foundry-QRT area model and is not counted as logic or as capacity saved by TSBG. The observed 58.13% reduction is bank activation, not deployed SRAM area. For the equal-bandwidth K8 comparison, adding that identical common capacity to both axes changes the K8/K1×8 area from 131,086/585,479 to 689,593/1,143,986 μm^2. The resulting foundry-QRT logic-plus-memory model retains a 39.72% area reduction and 1.687× directed-throughput/area advantage. It is an arithmetic model over matched capacity, not an integrated macro placement or routed timing result.

V. RELATED WORK

Prosperity already discovers subset/prefix parents, issues residual activations, and dependency-schedules their reconstruction [1]; C1 does not reclaim those ideas. Its circuits claim is their execution under a finite parent-product lifetime and one physical 1RW port, including grant-time liveness recheck, deadline arbitration, reserved response, forwarding, dead-write suppression, and atomic completion. Phi stores hierarchical patterns chosen by DSE [2]. We adopt their evaluation pattern—cycle simulation plus RTL/physical anchors and a baseline ladder—without relabeling their numbers as ours.

FireFly-T already broadcasts wider vectorized data from one memory bank across spatial-temporal work [8]; ELSA and SpikeX establish bundled Gustavson delivery and cross-window weight reuse [9], [10]. TSBG does not reclaim broadcast, bundling, group-major order, or weight reuse. Its specialization decides a row hit before the SRAM request and shares only that delivery among typed signed contexts while destination, tag, terminal, product, and Acc24 ownership remain private under the same cache/port contract. This direct prior also precludes comparing K8 against a single K1 lane.

SpinalFlow sorts compressed spike streams to preserve state locality, and SNE makes convolutional work proportional to the incoming event stream [4], [5]. Both establish event-driven execution. Neither removes repeated products under a finite 1RW parent lifetime or repeated weight delivery across private signed contexts. ODIN is a 28-nm digital neuromorphic core with online learning [3]; it is a silicon neuron/synapse fabric, not a product-capture convolution island. TCAS-II SRAM compute-in-memory briefs push sparsity into the array, including dynamic-sparsity control and PIM tutorials [6], [7]. C1 is a digital 1RW parent store with residual reconstruction, not an in-array MAC macro; we cite those briefs as the circuits-venue prior on sparsity, not as interchangeable PPA. A TCAS-II SNN accelerator combines weight-stationary/local-output-stationary reuse with address skipping and dynamic scheduling [11]; C2 differs by sharing a delivered row across polarity-bearing contexts whose products and accumulators must remain private. Another TCAS-II accelerator encodes inter-network weight deltas to reduce external-memory access [12]. That static representation is orthogonal to TSBG: our weights are unchanged, and the saved event is a runtime read of the same row by multiple live contexts. The comparison reinforces why request count, logic cost, and SRAM energy must be reported separately.

A 28-nm optical-flow chip jointly reports operations, external-memory traffic, latency, energy, and AEE [13]. We follow that separation of metrics and do not numerically

TABLE V
DIRECT-PRIOR BOUNDARY; INHERITED MECHANISMS ARE NOT RECLAIMED.

Prior	Inherited mechanism	Circuit specialization here
Prosperity	subset parent, residual, dependency	finite lifetime; one 1RW grant; atomic completion
FireFly-T / ELSA	bank broadcast; bundled Gustavson delivery	typed sign/destination/Acc24 private; hit before read request
Phi	hierarchical sparse pattern and DSE	no pattern claim; physical port/capacity ladder
WS-LOS	weight/local-output reuse and skipping	same-cache token-set A/B with private products
Delta weight sharing	static representation and traffic reduction	unchanged weights; runtime pre-read context broadcast

rank its silicon measurements against this prelayout island. For a co-designed network, a published chip rarely supports the identical graph. Comparison therefore stays at three levels: same-network component baselines, iso-workload public-artifact opportunities, and published technology-normalized metrics. Results from different levels are never multiplied into a system speedup.

VI. Scope and Limitations

C1 and C2 remove different max terms. C1 reuses a signed product while paying for finite parent lifetime and one 1RW grant; C2/TSBG reuses a weight delivery while retaining private products and accumulators. Their ratios are therefore not multiplied. On the C1 ledger, same-coordinate bit skipping gives only 1.003×, whereas exact parent reuse gives 1.6945×; the remaining 12.25% gap to the concurrent-access ceiling is the measured single-port tax.

In the 2,880-workload direct-VCS population, TSBG has seven slightly slower nonempty cases because group-major order can lose ordinary-cache residency under weak reuse. These cases remain in the ratio of sums. The reported population spans every FC1/FC2 layer identity but uses three fixed B4 regions per layer/sample; it does not imply full-FC wall time. Ordinary scheduling remains an exact fallback.

C1's mapped nine-SRAM island and full 214,912-B ledger are separate evidence; the 105-macro capacity is a 0.988-mm^2 area model rather than integrated PPA. Its 22.07-nJ result is a directed prelayout window, not frame energy. C2/TSBG is presently setup-clean, but hold and power remain open; its common 288-KiB weight store is modeled separately. No component ratio is promoted to monolithic FPS, energy/frame, or whole-network speedup.

VII. Conclusion

Exact reuse requires matching data lifetime to physical service. C1 combines finite parent residency with single-port reconstruction; C2/TSBG combines shared weight delivery with private signed accumulation. Their evaluation separates modeled product-capture latency, RTL-observed read suppression, and 28-nm circuit cost. The resulting component design targets redundant transfers that remain after ordinary zero-source skipping, without changing the evaluated source arithmetic.

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